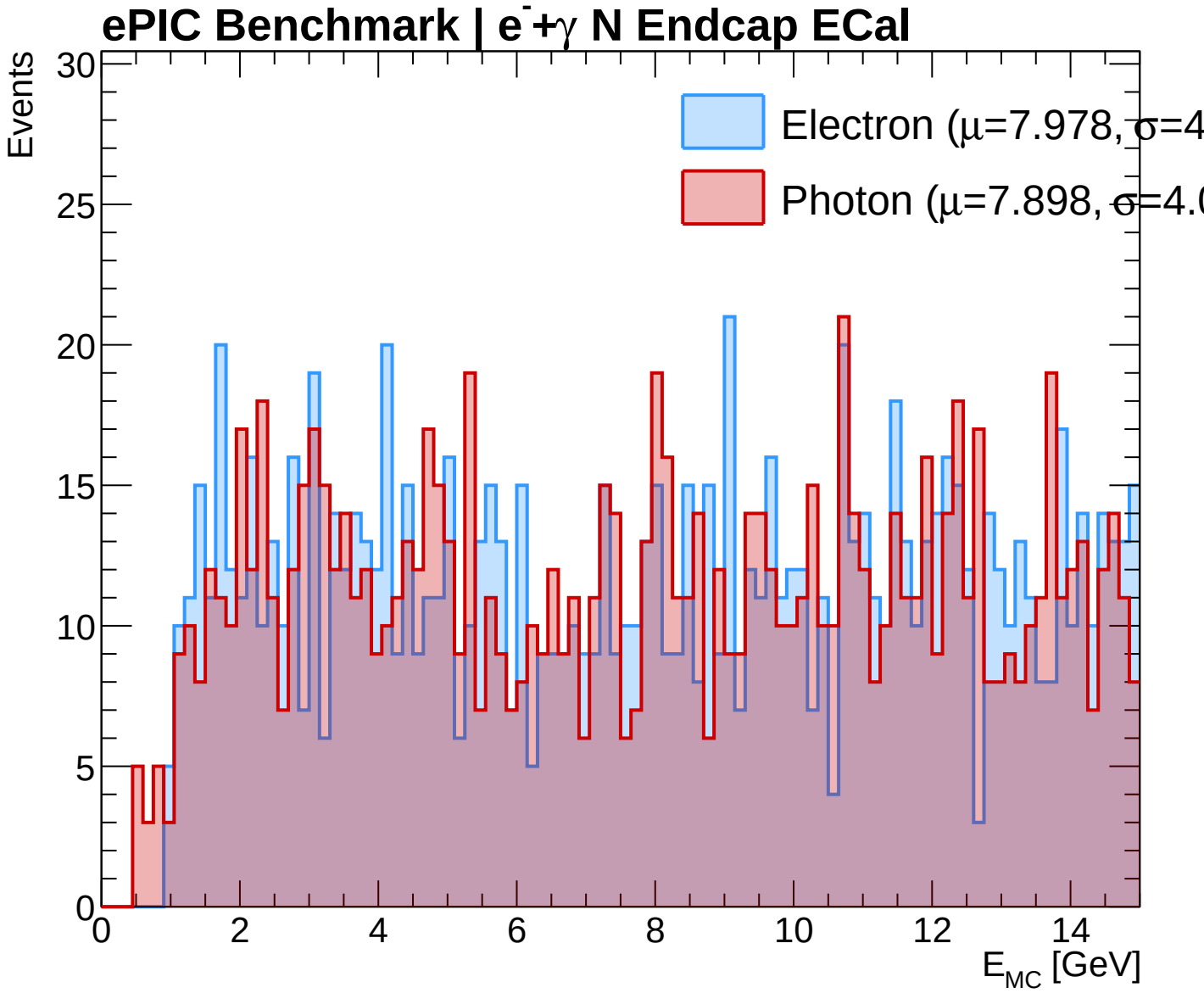
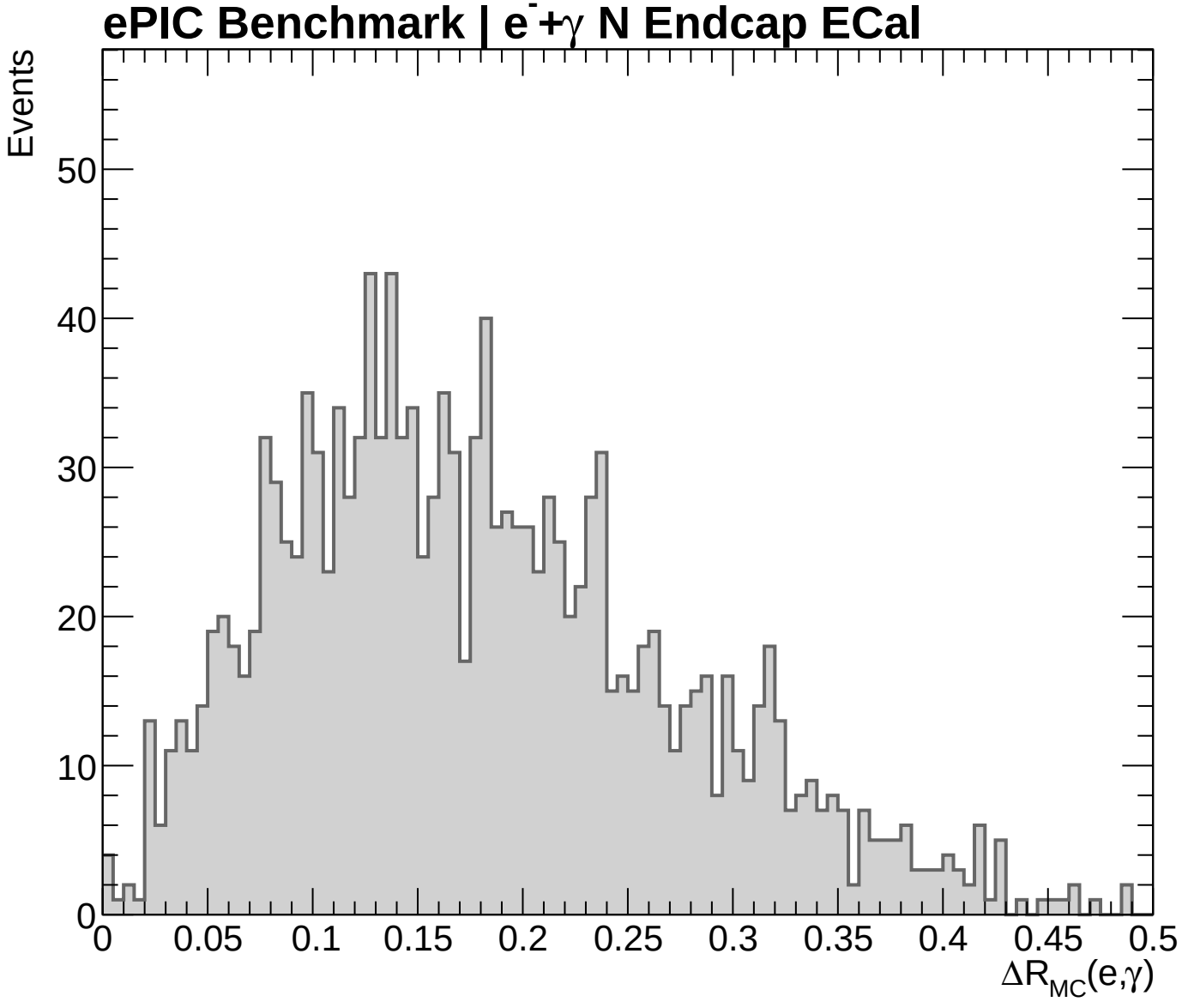
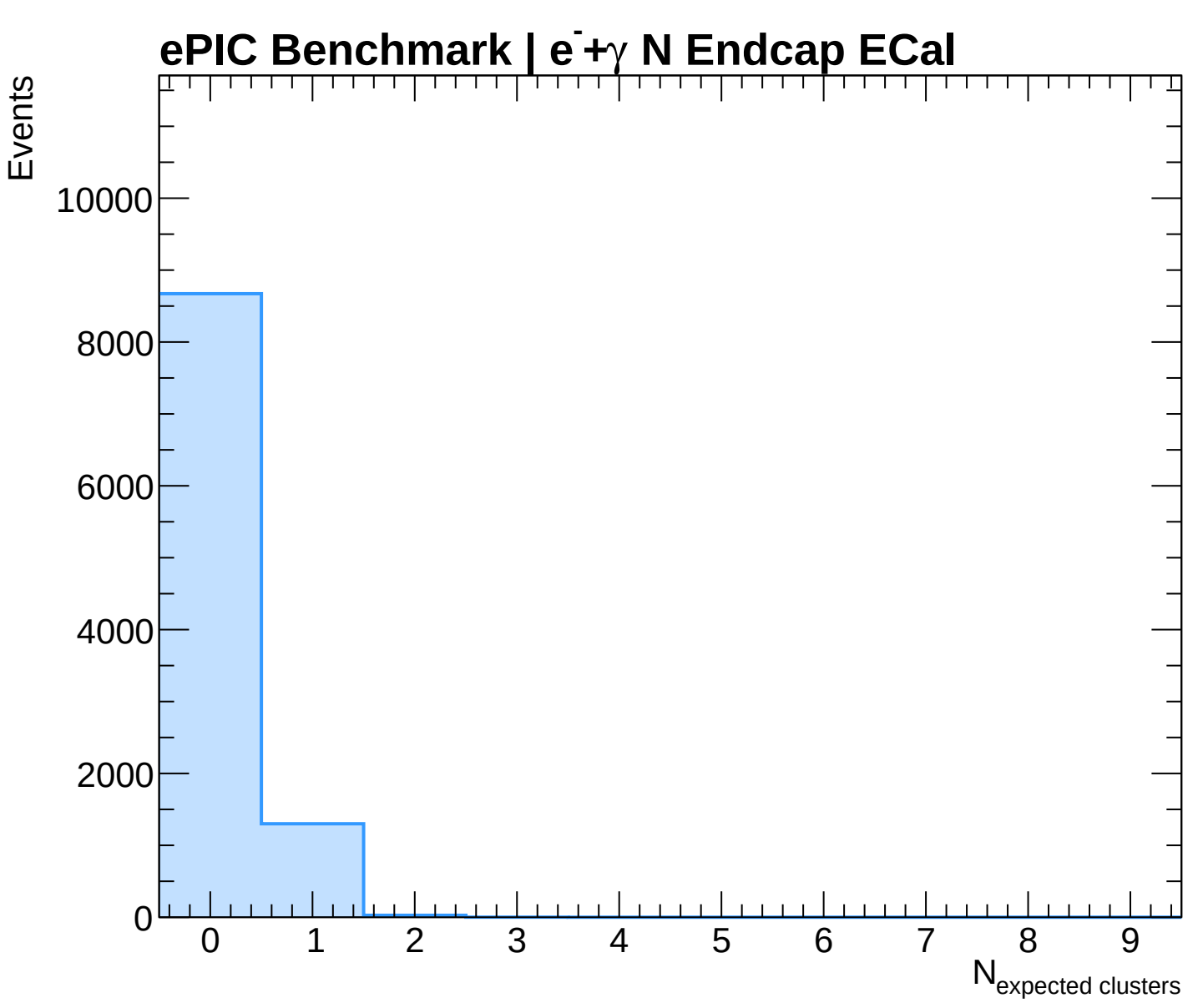
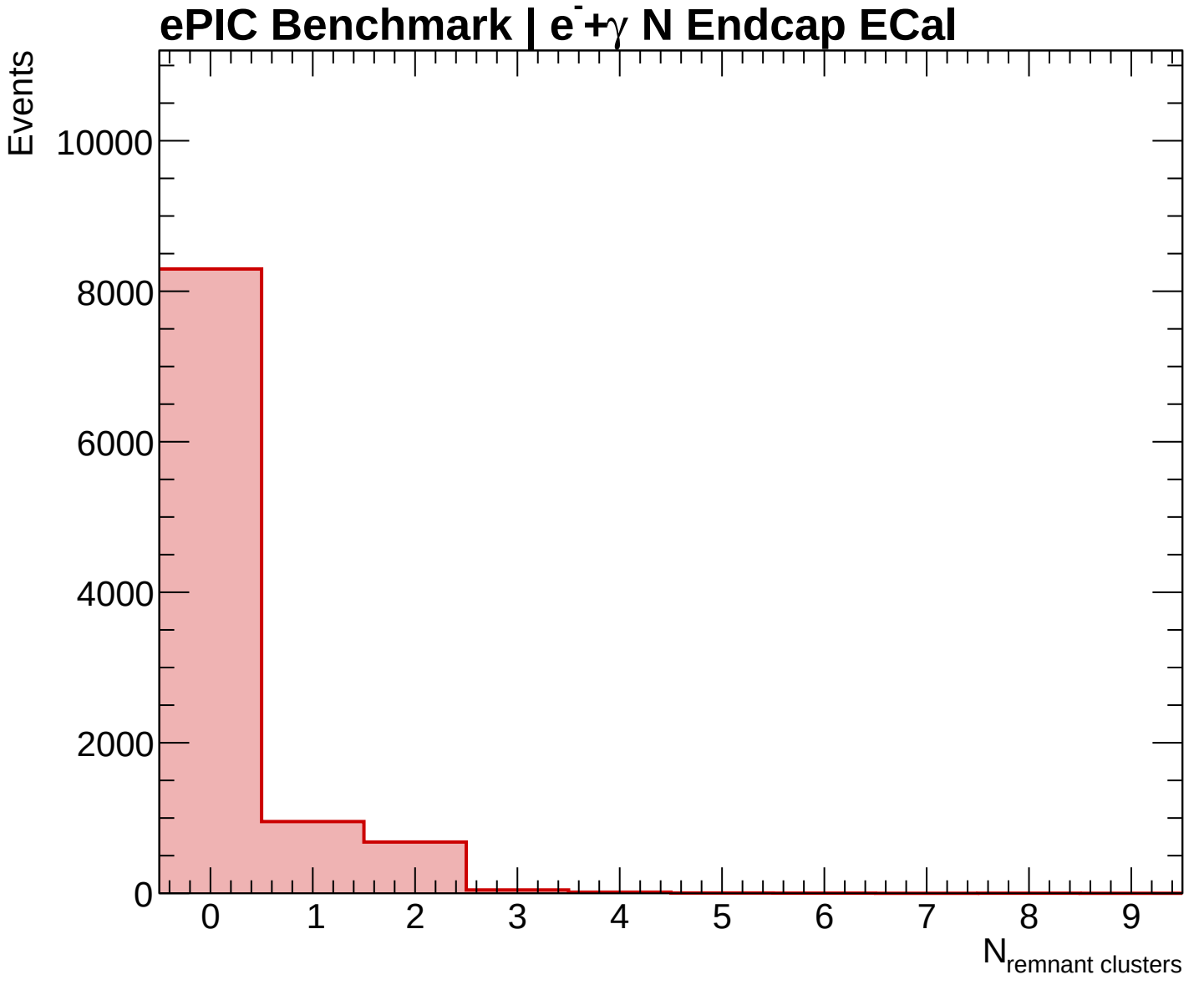
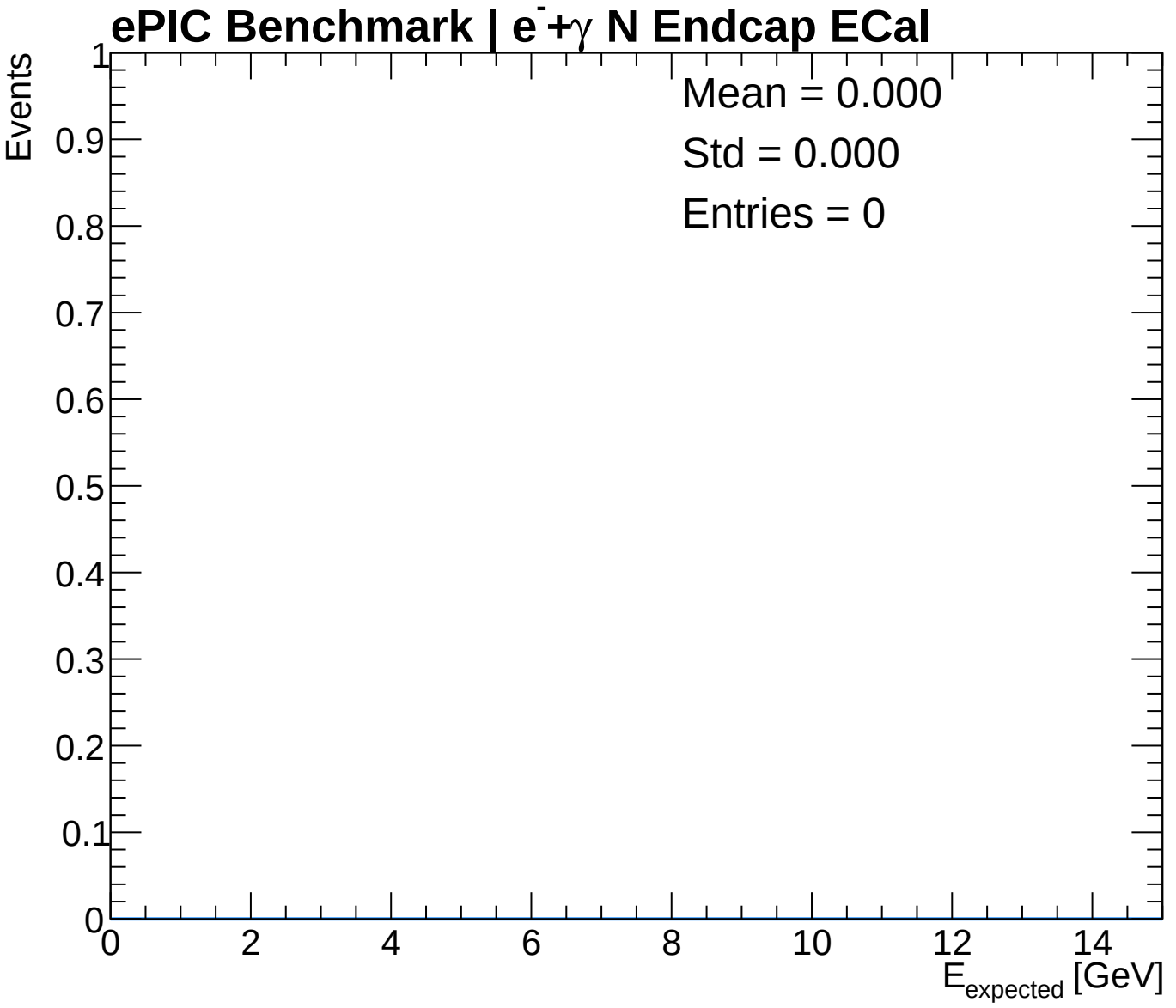
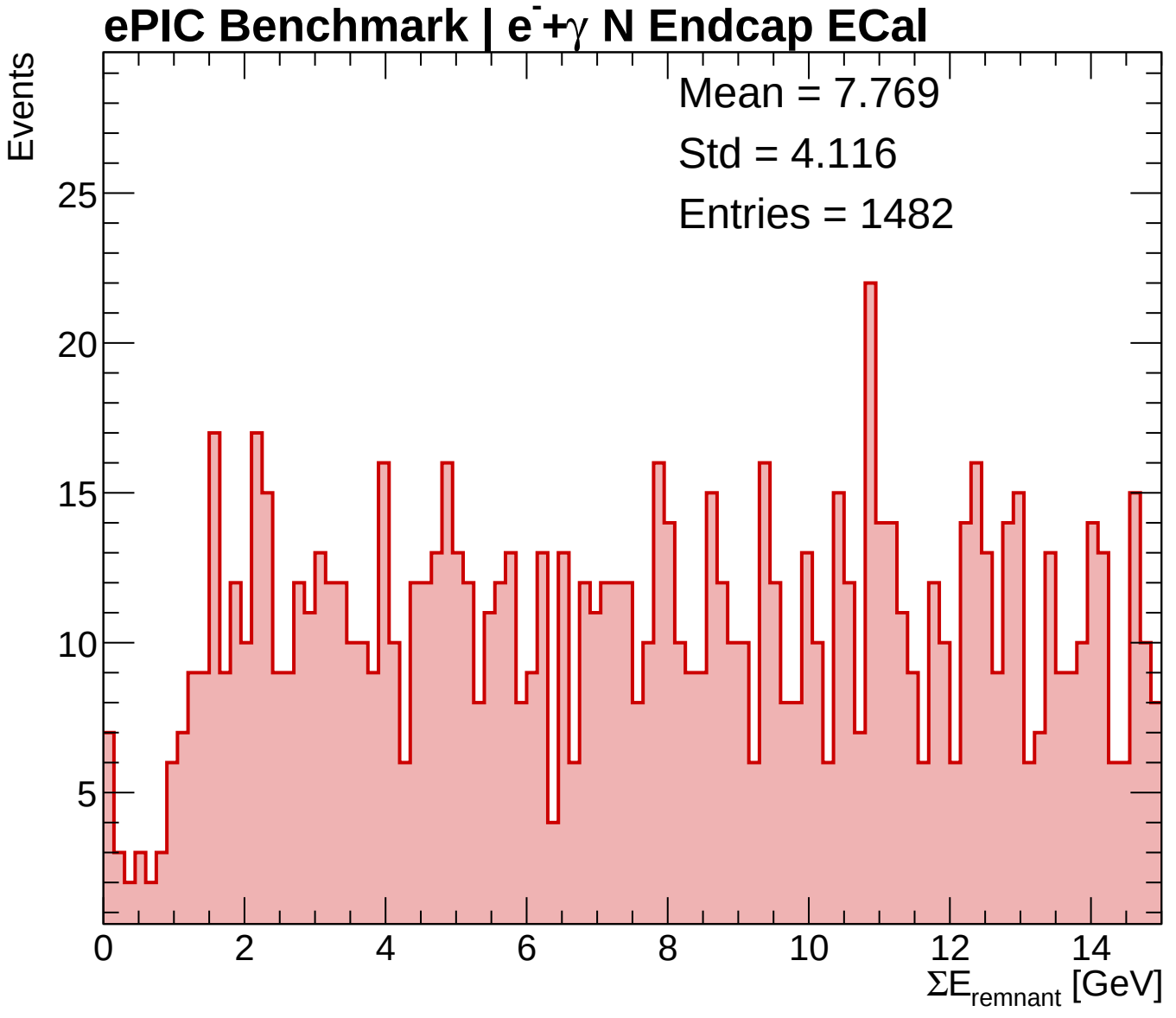

ePIC Benchmark | e⁻+γ N Endcap ECal
Events
Electron (μ=7.978, σ=4
Photon (μ=7.898, σ=4.0
0
5
10
15
20
25
30
0
2
4
6
8
10
12
14
E_MC [GeV]

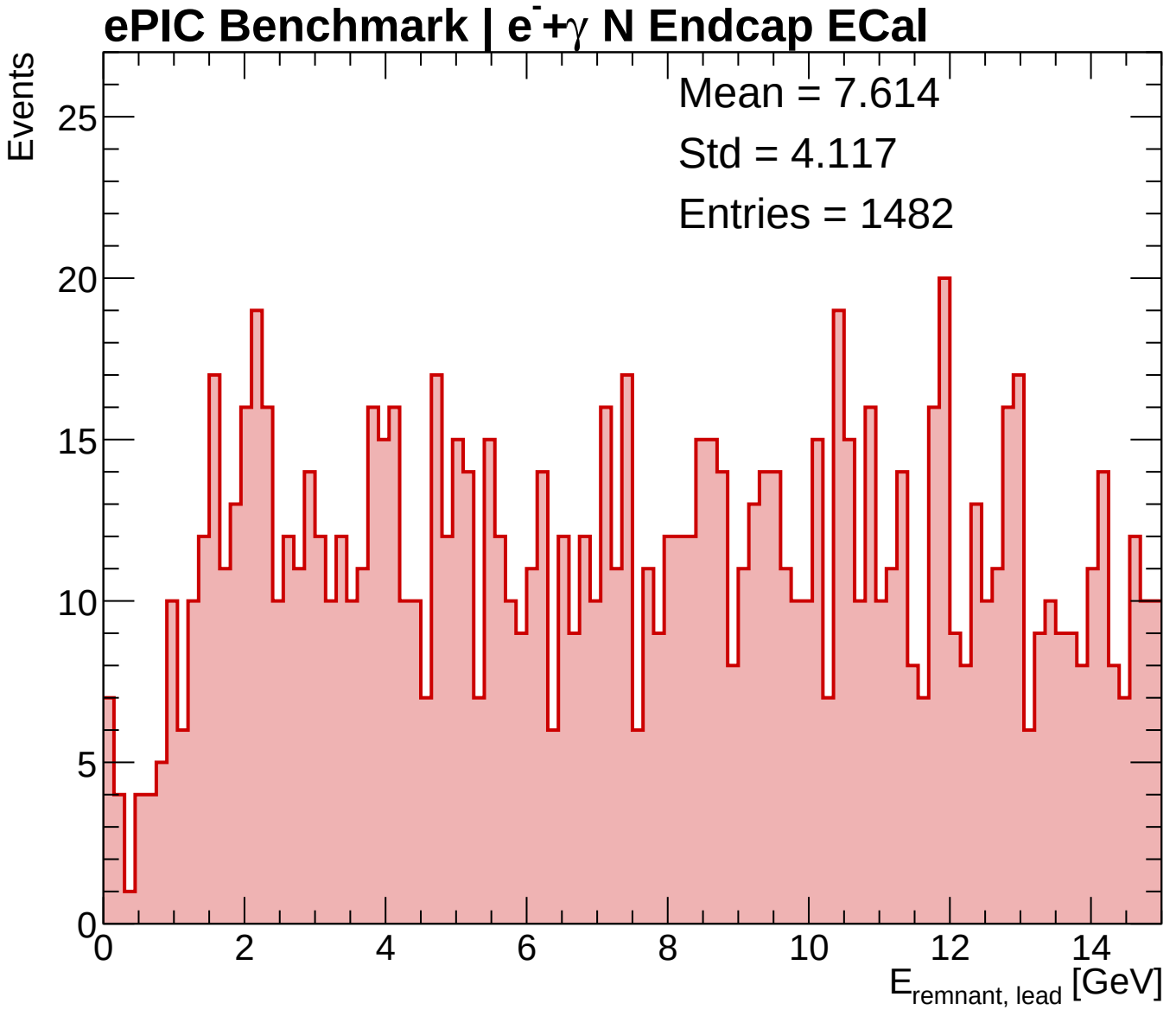

ePIC Benchmark | e⁻+γ N Endcap ECal
Events
0
10
20
30
40
50
0
0.05
0.1
0.15
0.2
0.25
0.3
0.35
0.4
0.45
0.5
ΔR_MC(e,γ)

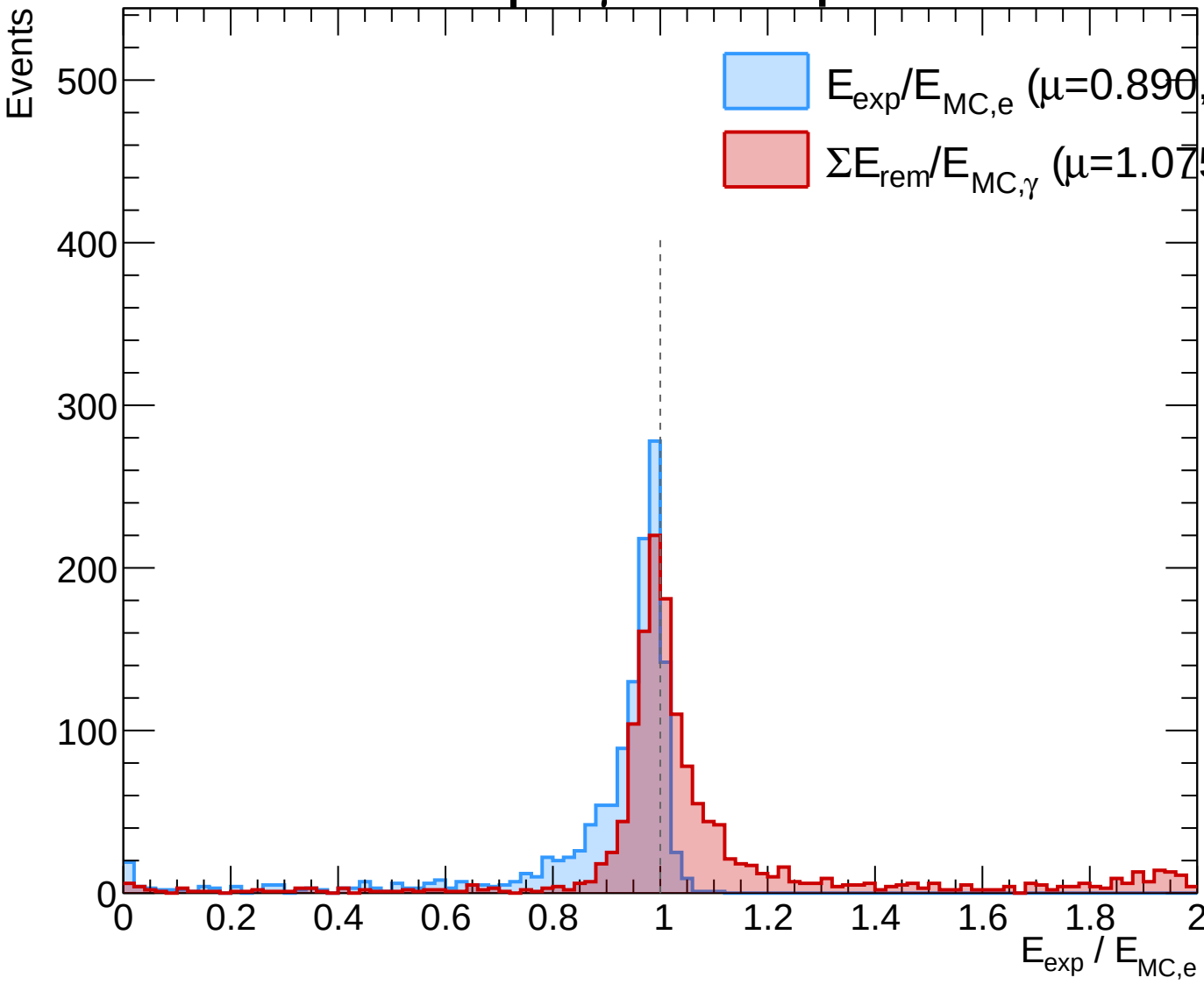

ePIC Benchmark | e⁻+γ N Endcap ECal
Events
10000
8000
6000
4000
2000
0
0
1
2
3
4
5
6
7
8
9
N expected clusters

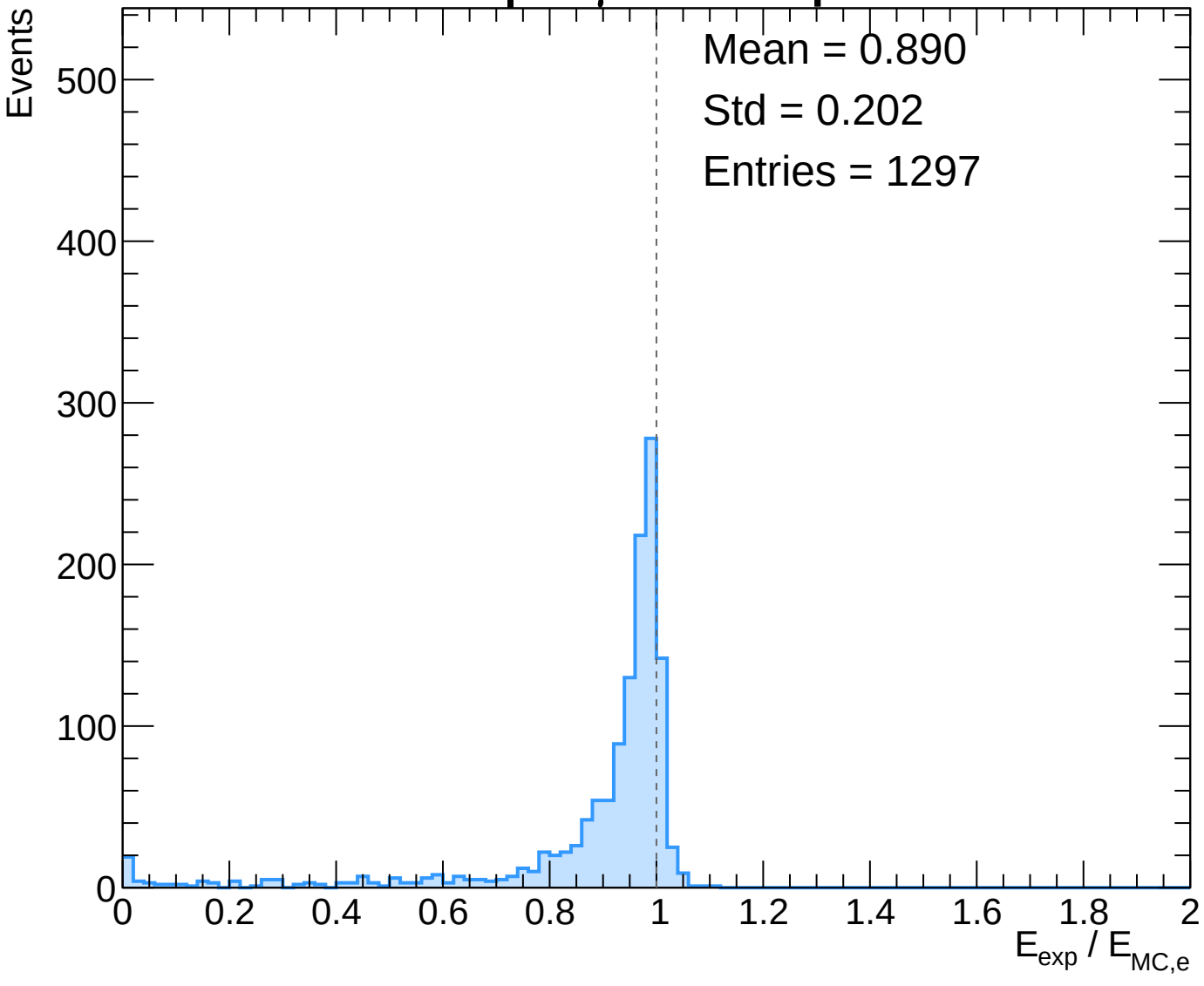

ePIC Benchmark | e⁻+γ N Endcap ECal
Events
10000
8000
6000
4000
2000
0
0
1
2
3
4
5
6
7
8
9
N remnant clusters

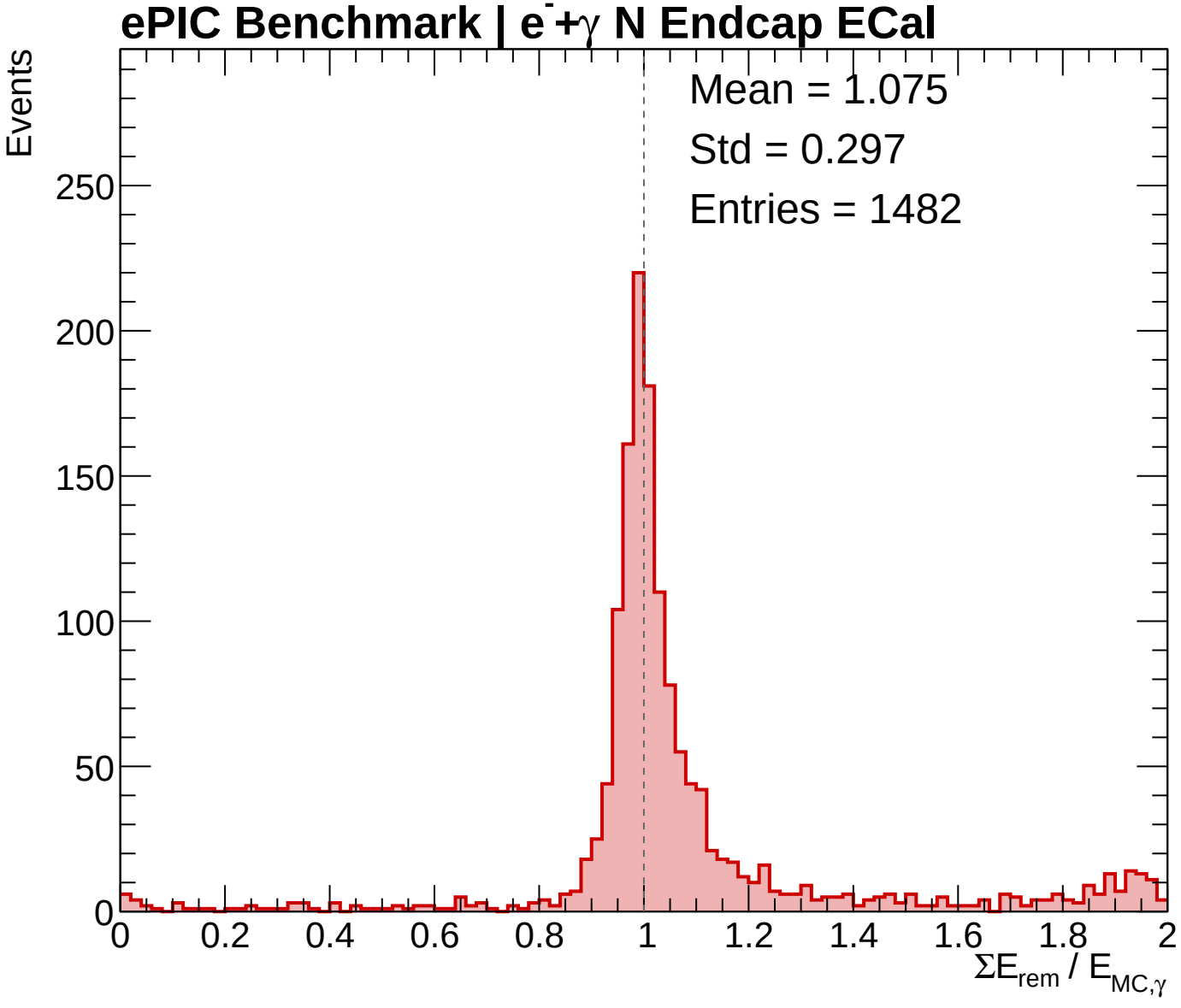

ePIC Benchmark | e⁻+γ N Endcap ECal
Events
Mean = 0.000
Std = 0.000
Entries = 0
1
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0
0
2
4
6
8
10
12
14
E_expected [GeV]

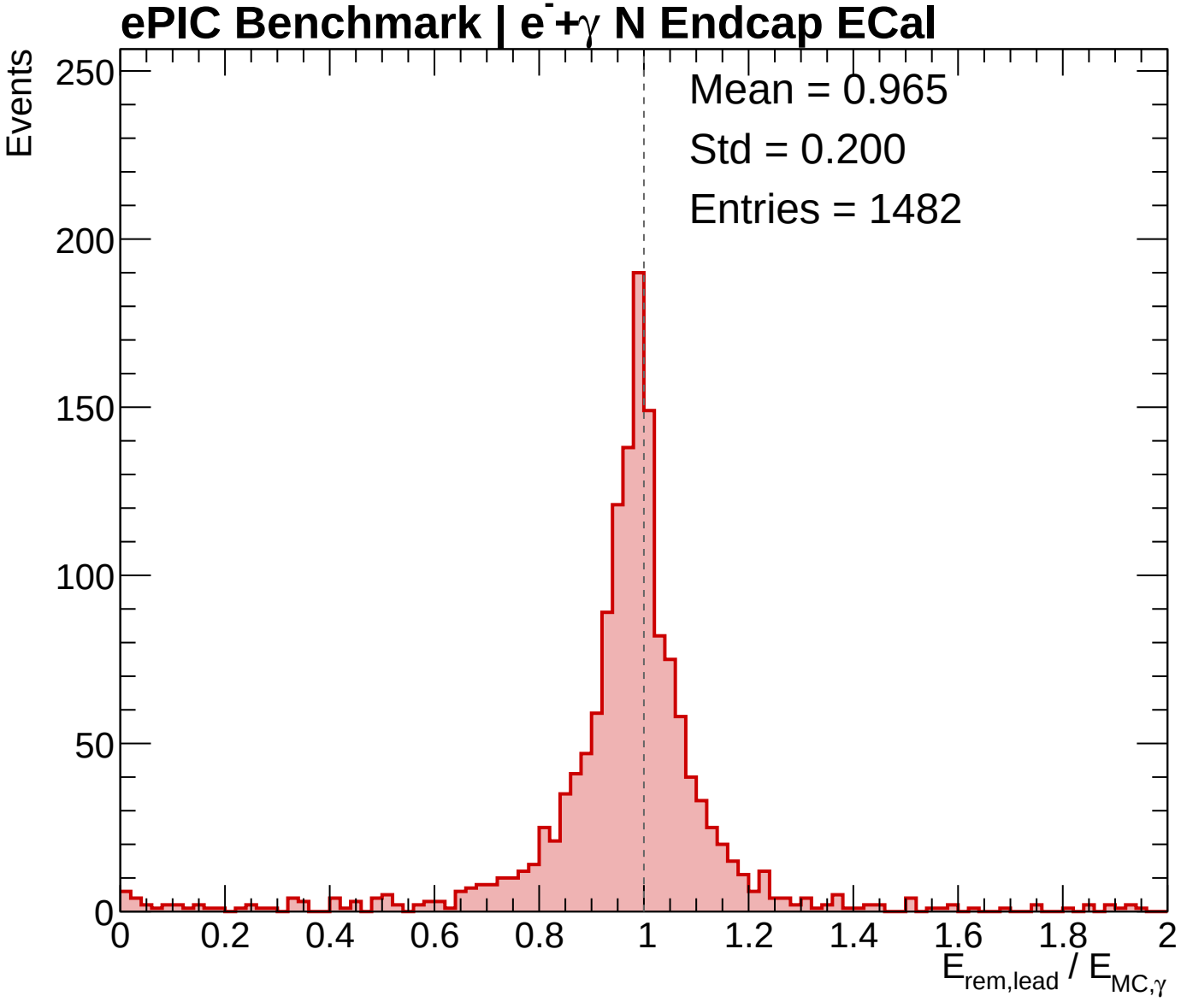

ePIC Benchmark | e⁻+γ N Endcap ECal
Mean = 7.769
Std = 4.116
Entries = 1482
Events
25
20
15
10
5
0
2
4
6
8
10
12
14
ΣE_remnant [GeV]

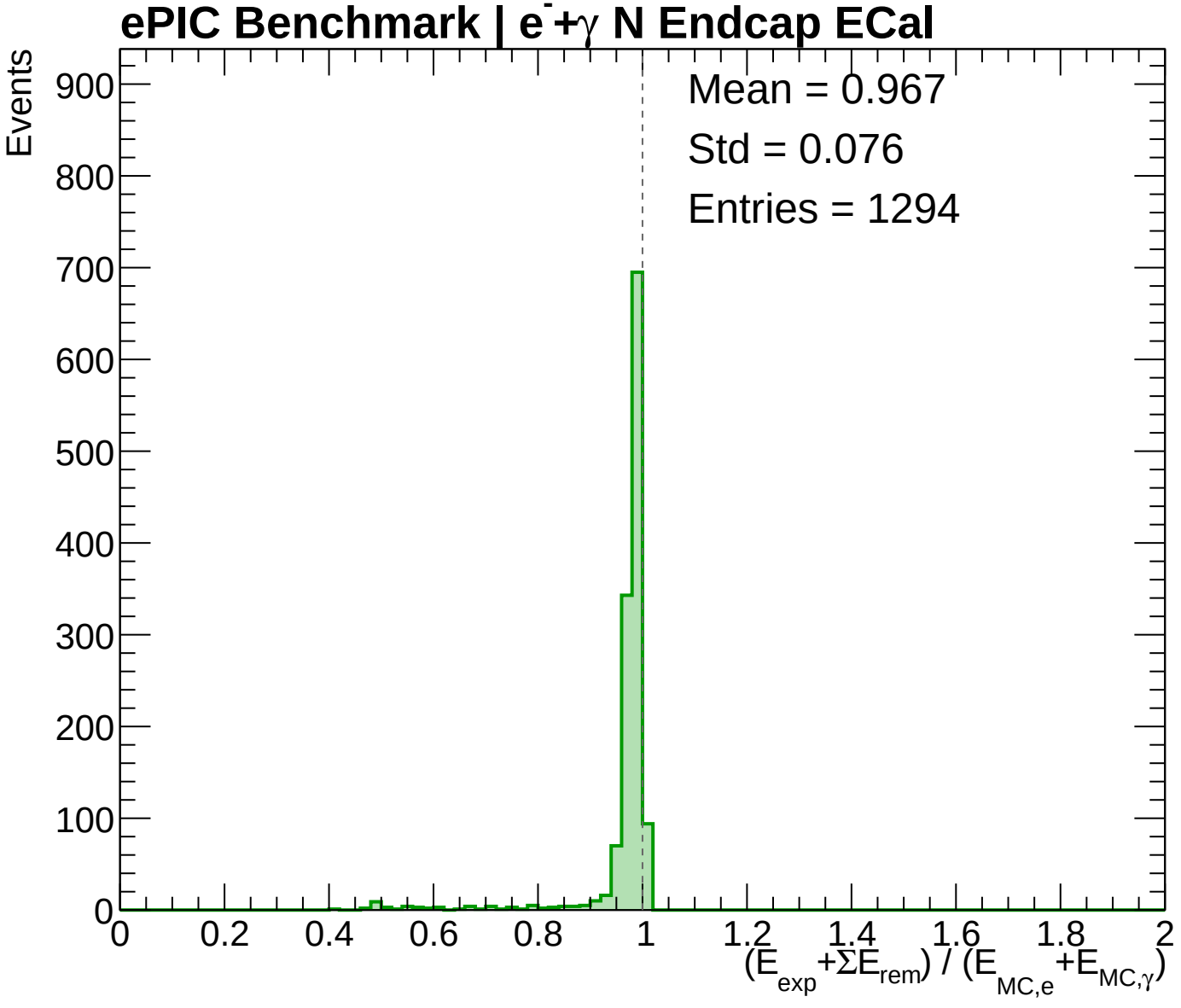

ePIC Benchmark | e⁻+γ N Endcap ECal
Mean = 7.614
Std = 4.117
Entries = 1482
Events
0
5
10
15
20
25
0
2
4
6
8
10
12
14
E remnant, lead [GeV]

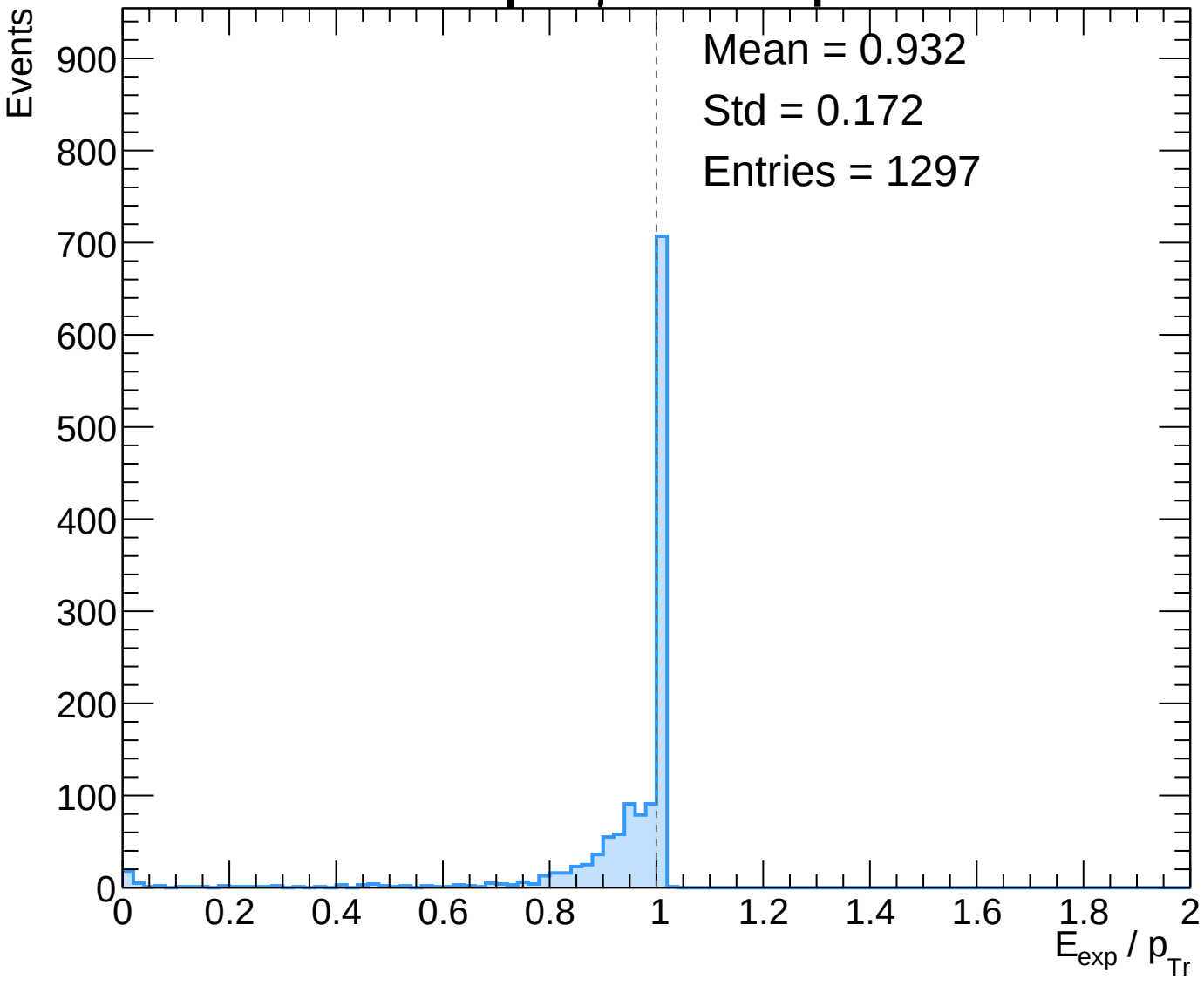

ePIC Benchmark | e⁻+γ N Endcap ECal
Events
500
400
300
200
100
0
0
0.2
0.4
0.6
0.8
1
1.2
1.4
1.6
1.8
2
E_exp / E_MC,e
E_exp/E_MC,e (μ=0.890, σ=
ΣE_rem/E_MC,γ (μ=1.075, σ

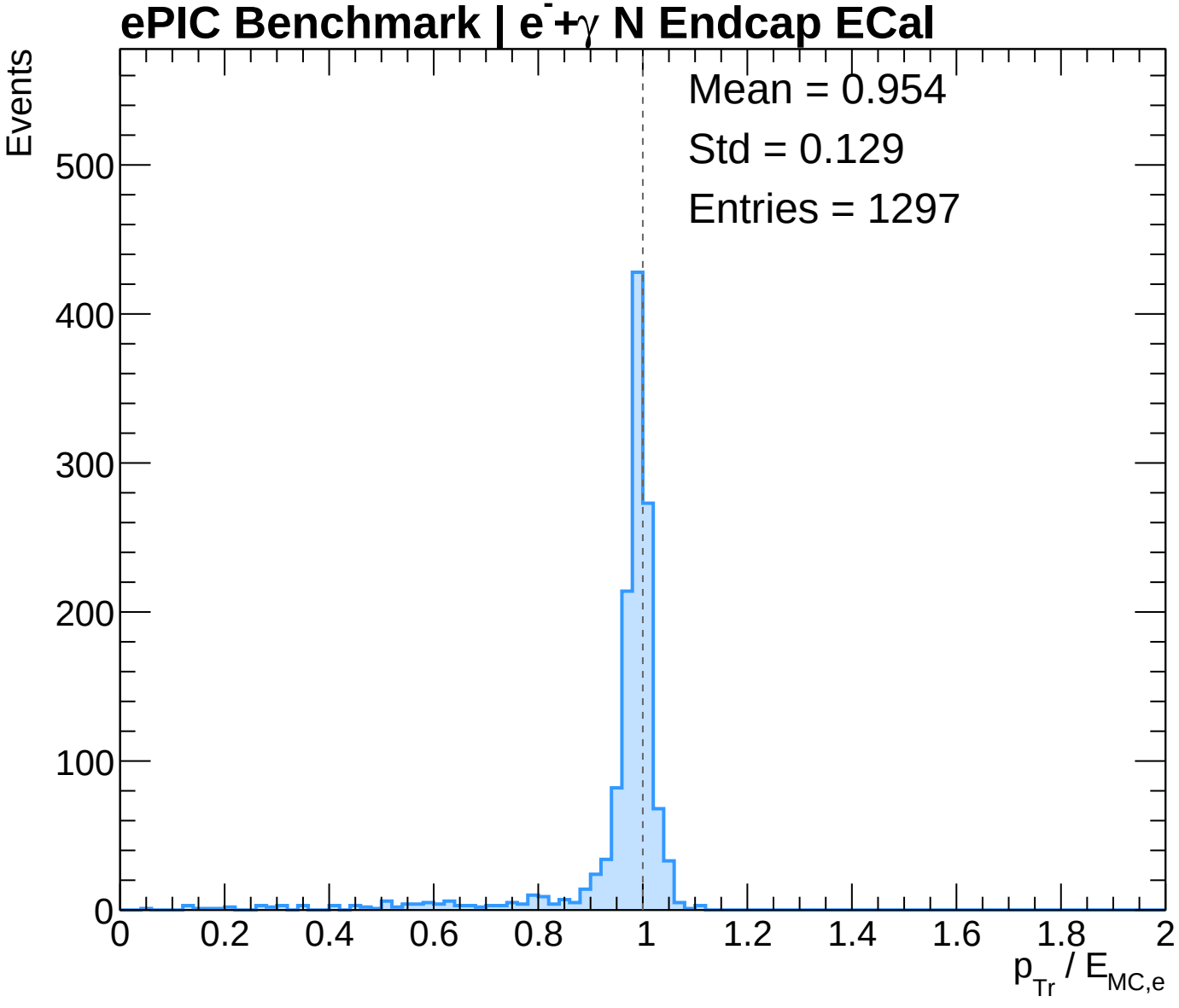

ePIC Benchmark | e⁻+γ N Endcap ECal
Events
Mean = 0.890
Std = 0.202
Entries = 1297
0
100
200
300
400
500
0
0.2
0.4
0.6
0.8
1
1.2
1.4
1.6
1.8
2
E_exp / E_MC,e

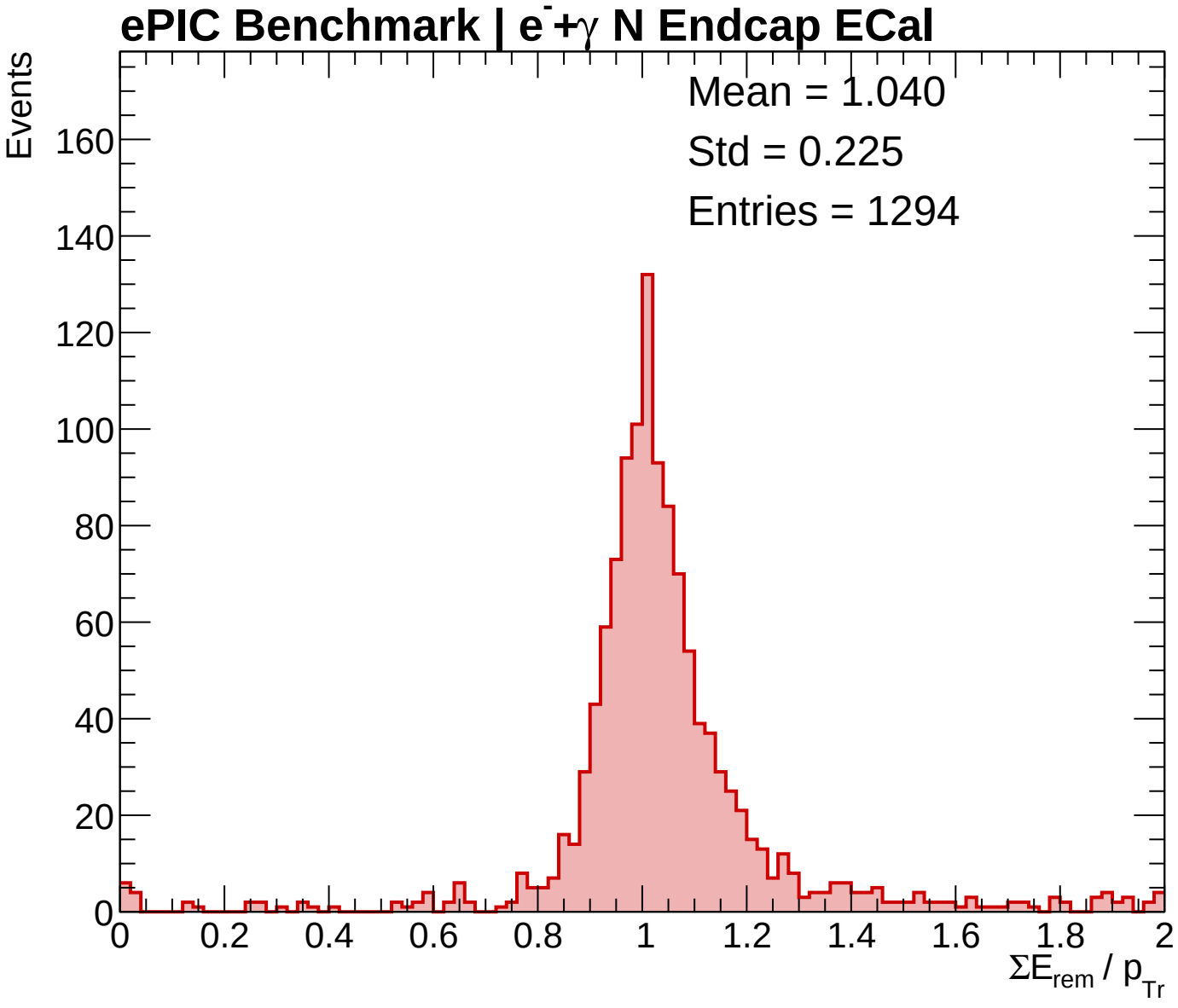

ePIC Benchmark | e⁻+γ N Endcap ECal
Events
Mean = 1.075
Std = 0.297
Entries = 1482
0
50
100
150
200
250
0
0.2
0.4
0.6
0.8
1
1.2
1.4
1.6
1.8
2
ΣE_rem / E_MC,γ

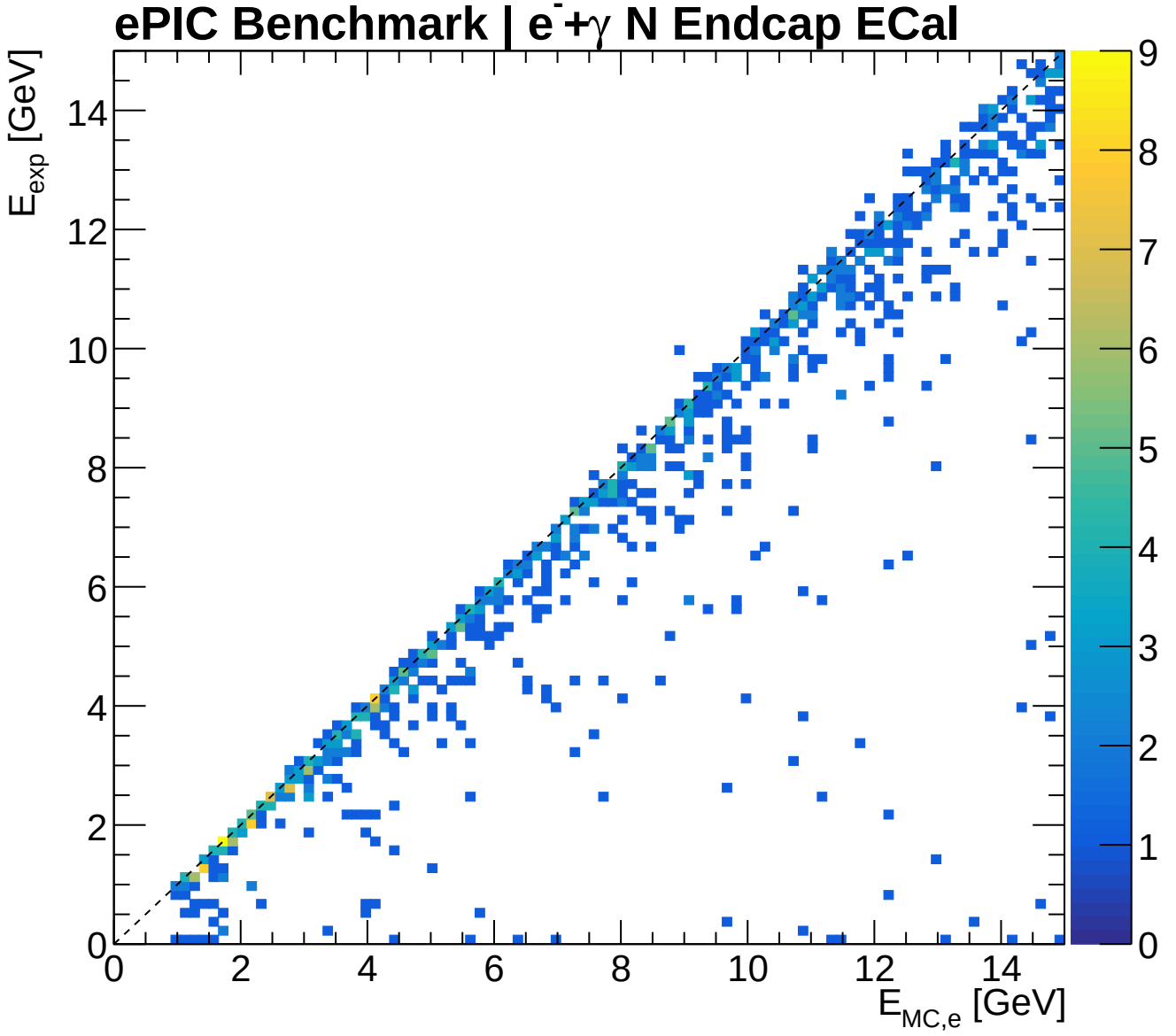

ePIC Benchmark | e⁻+γ N Endcap ECal
Events
Mean = 0.965
Std = 0.200
Entries = 1482
250
200
150
100
50
0
0
0.2
0.4
0.6
0.8
1
1.2
1.4
1.6
1.8
2
E_rem,lead / E_MC,γ

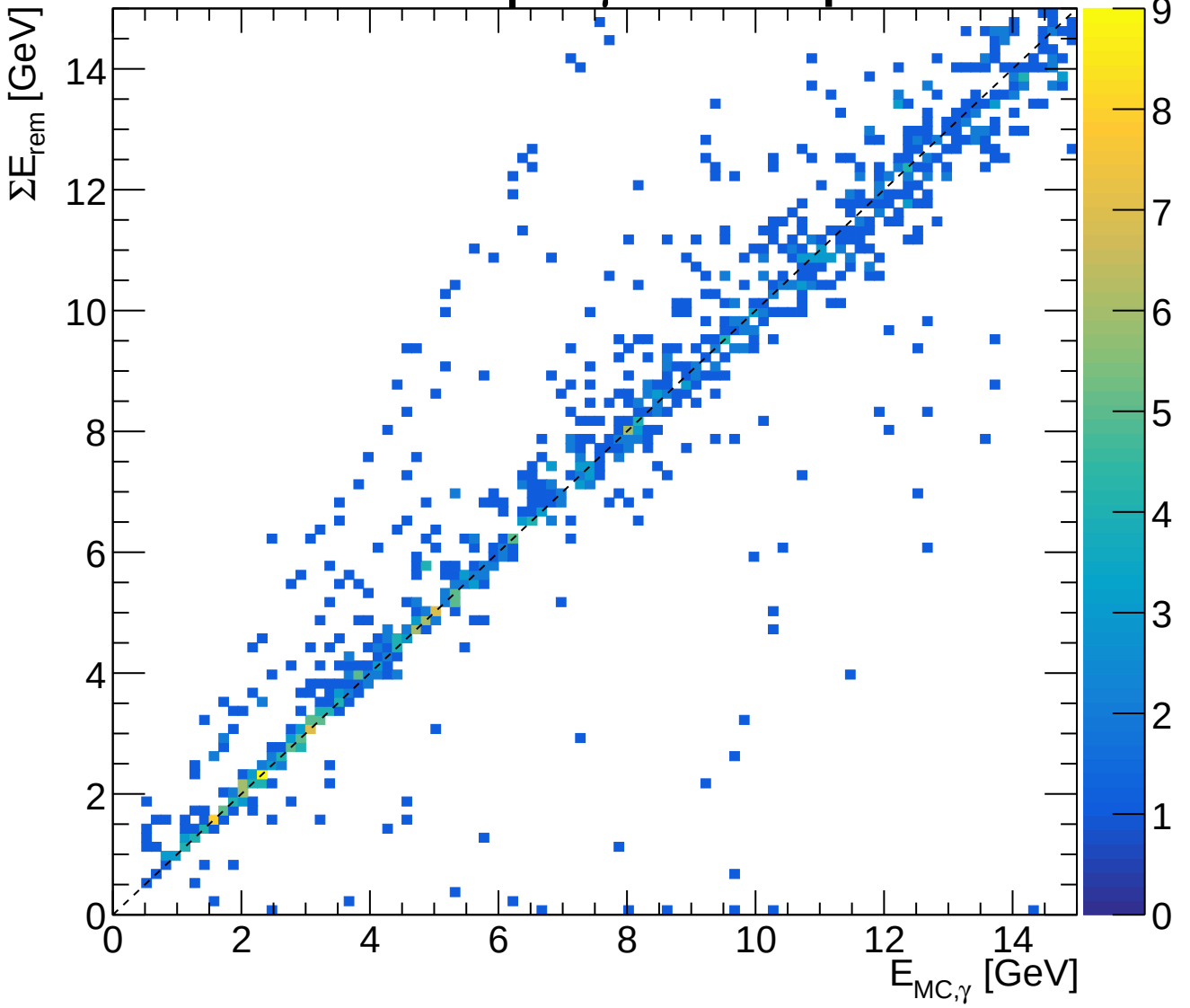

ePIC Benchmark | e⁻+γ N Endcap ECal
Events
Mean = 0.967
Std = 0.076
Entries = 1294
0
100
200
300
400
500
600
700
800
900
0
0.2
0.4
0.6
0.8
1
1.2
1.4
1.6
1.8
2
(E_exp+ΣE_rem) / (E_MC,e+E_MC,γ)

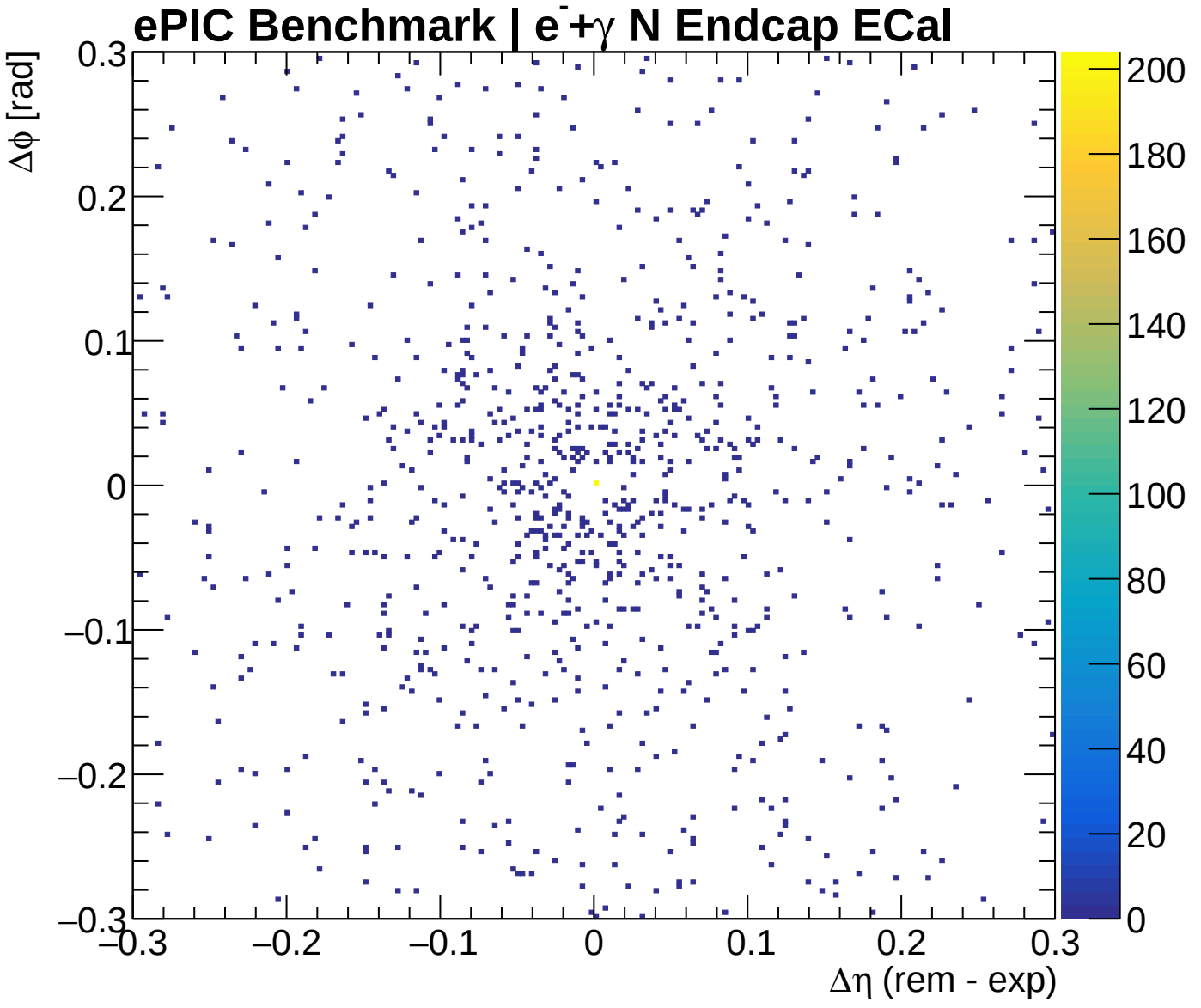

ePIC Benchmark | e⁻+γ N Endcap ECal
Events
Mean = 0.932
Std = 0.172
Entries = 1297
0
100
200
300
400
500
600
700
800
900
0
0.2
0.4
0.6
0.8
1
1.2
1.4
1.6
1.8
2
E_exp / p_Tr

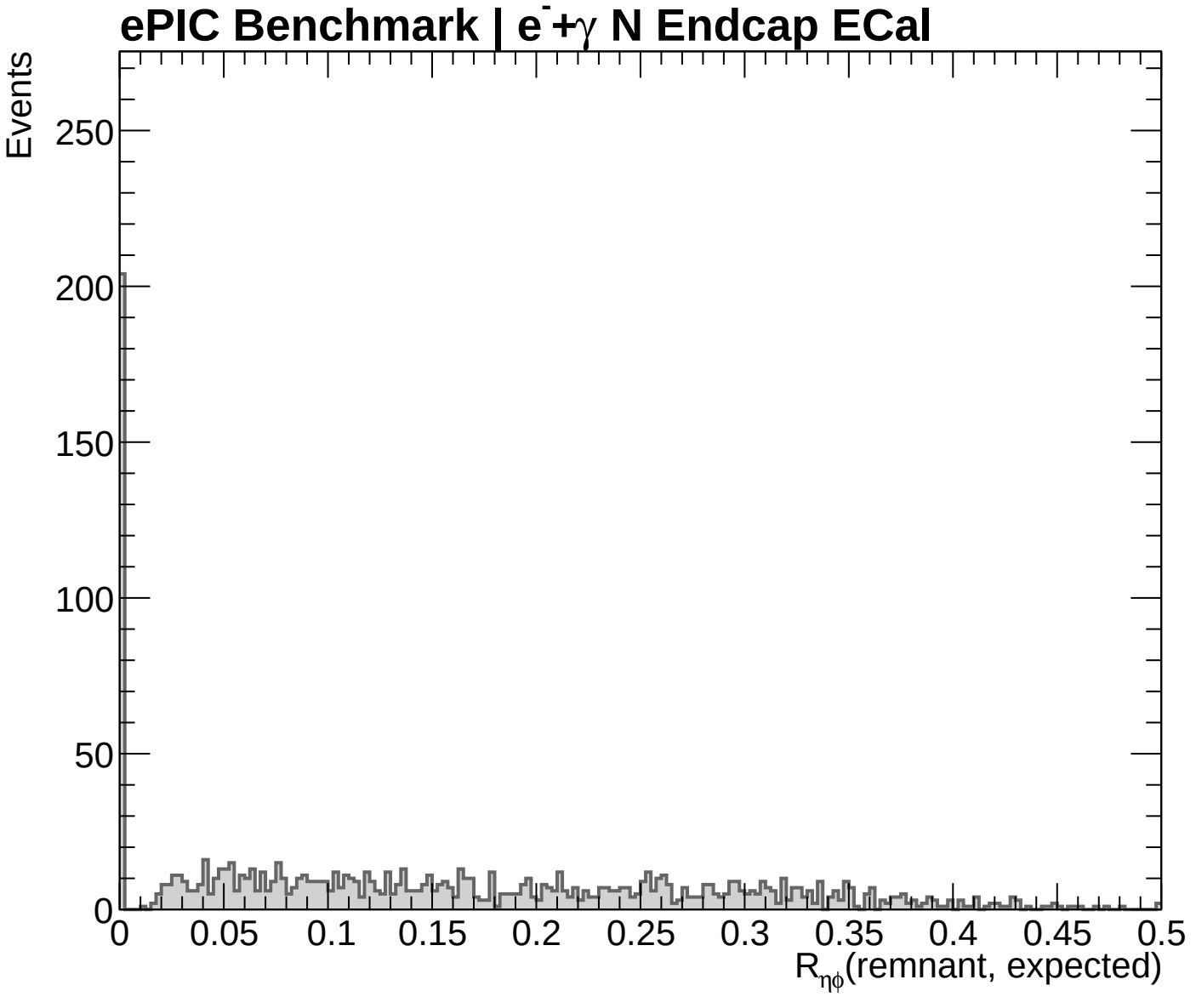

ePIC Benchmark | e⁻+γ N Endcap ECal
Events
Mean = 0.954
Std = 0.129
Entries = 1297
0
100
200
300
400
500
0
0.2
0.4
0.6
0.8
1
1.2
1.4
1.6
1.8
2
p_Tr / E_MC,e

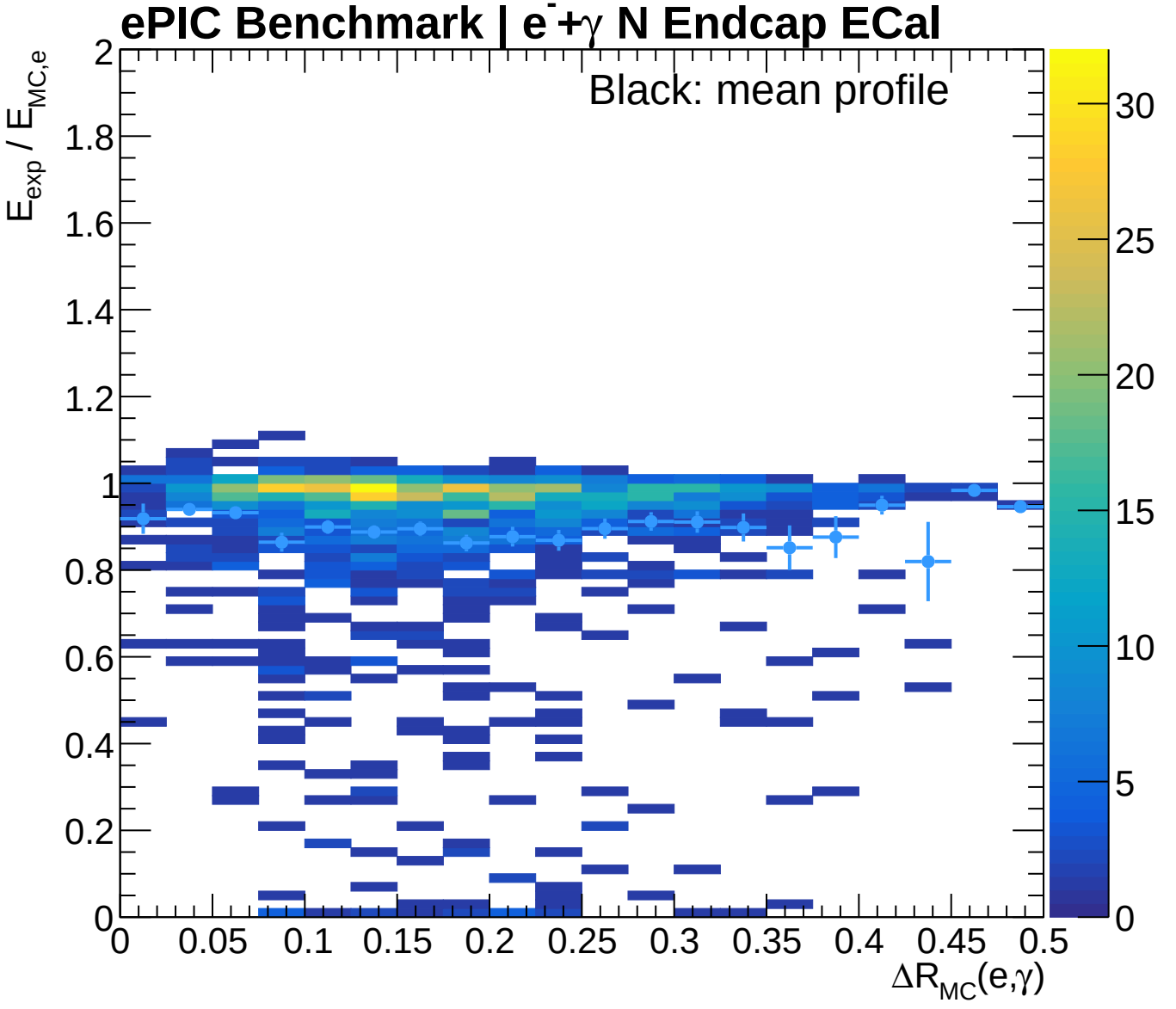

ePIC Benchmark | e⁻+γ N Endcap ECal
Events
Mean = 1.040
Std = 0.225
Entries = 1294
0
20
40
60
80
100
120
140
160
0
0.2
0.4
0.6
0.8
1
1.2
1.4
1.6
1.8
2
ΣE_rem / p_Tr

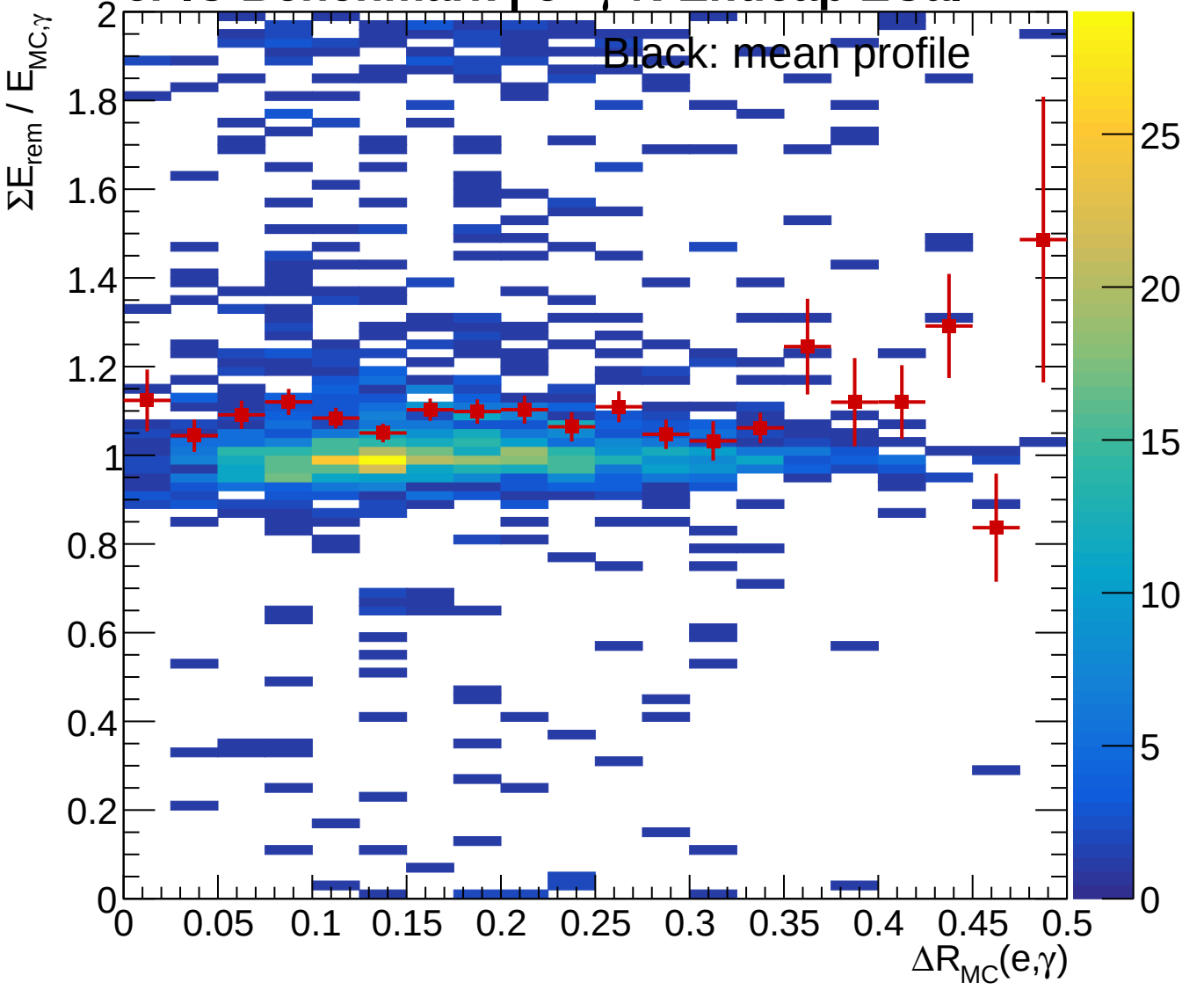

ePIC Benchmark | e⁻+γ N Endcap ECal
E_exp [GeV]
E_MC,e [GeV]
0
2
4
6
8
10
12
14
0
1
2
3
4
5
6
7
8
9

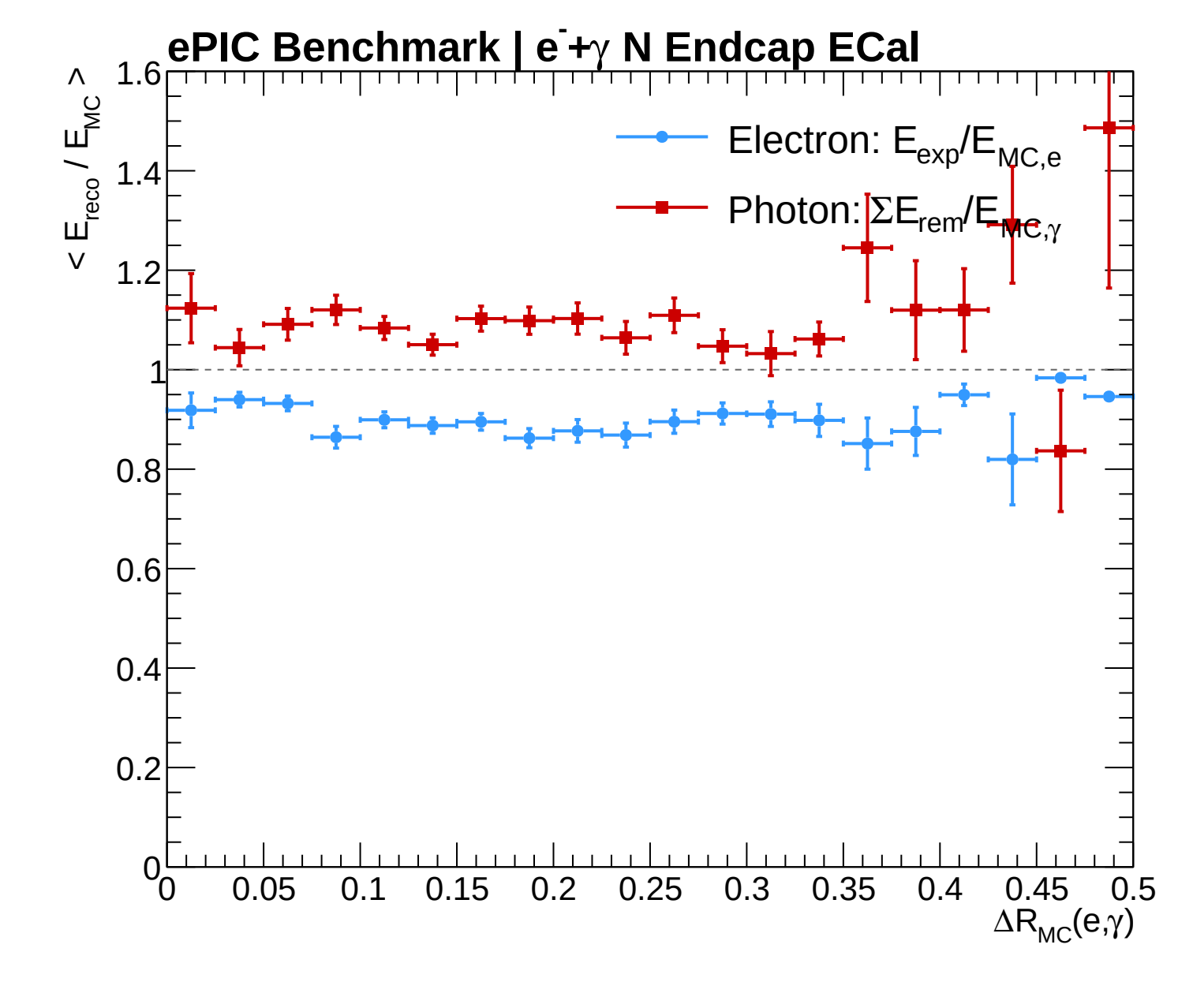

ePIC Benchmark | e⁻+γ N Endcap ECal
ΣE_rem [GeV]
E_MC,γ [GeV]
0
2
4
6
8
10
12
14
0
1
2
3
4
5
6
7
8
9

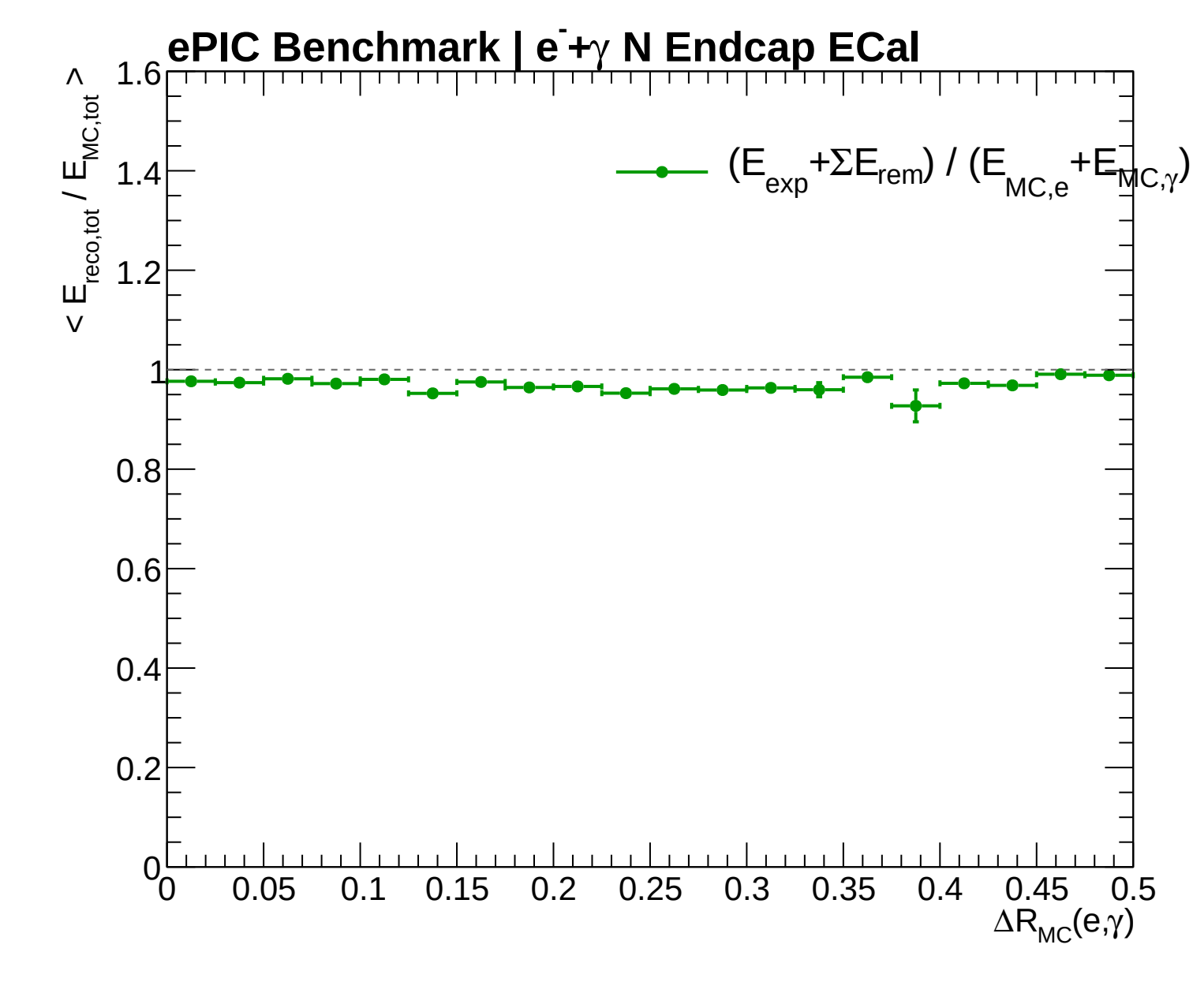

ePIC Benchmark | e⁻+γ N Endcap ECal
Δϕ [rad]
0.3
0.2
0.1
0
−0.1
−0.2
−0.3
−0.3
−0.2
−0.1
0
0.1
0.2
0.3
Δη (rem - exp)
200
180
160
140
120
100
80
60
40
20
0

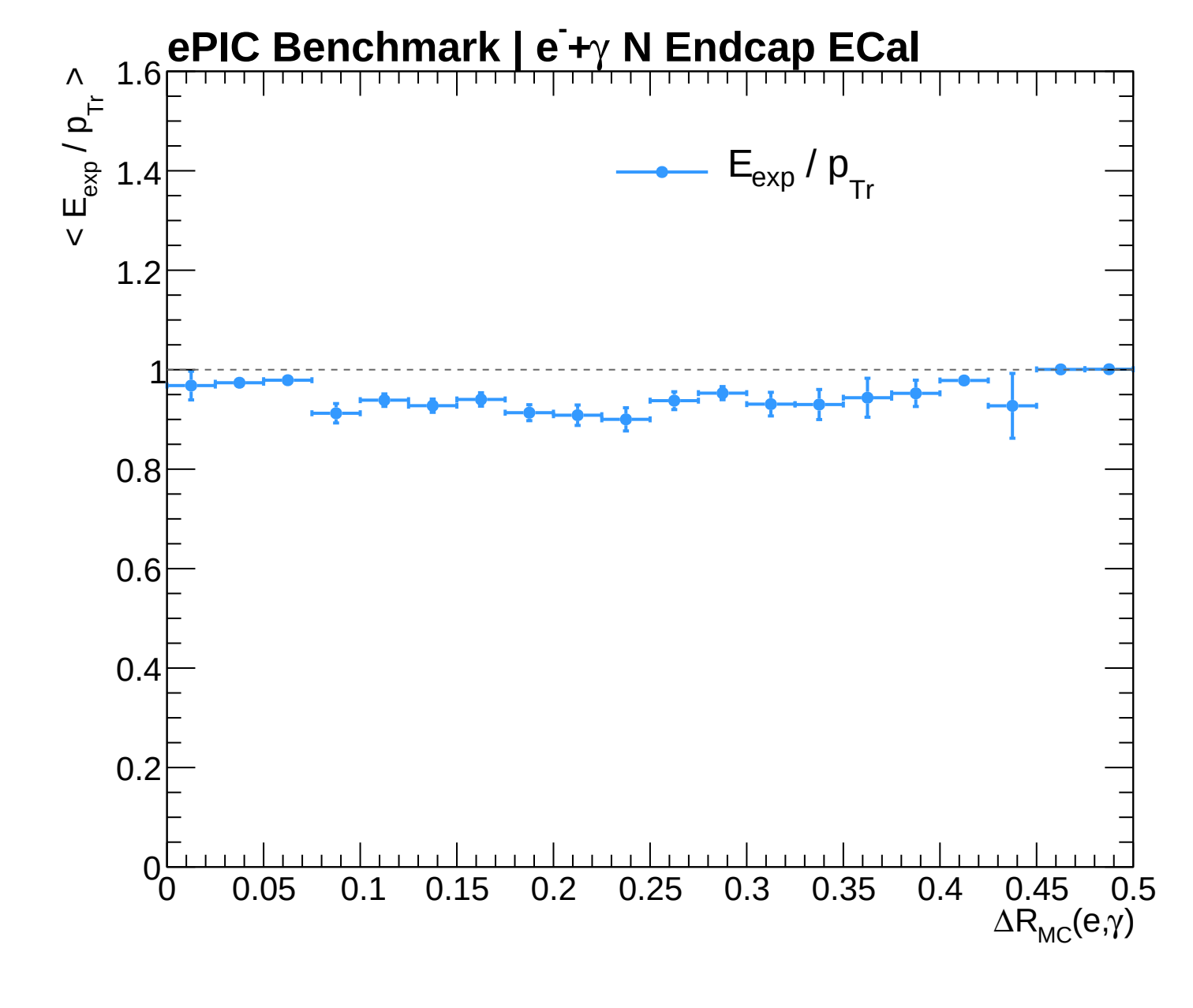

ePIC Benchmark | e⁻+γ N Endcap ECal
Events
250
200
150
100
50
0
0
0.05
0.1
0.15
0.2
0.25
0.3
0.35
0.4
0.45
0.5
R_ηφ(remnant, expected)

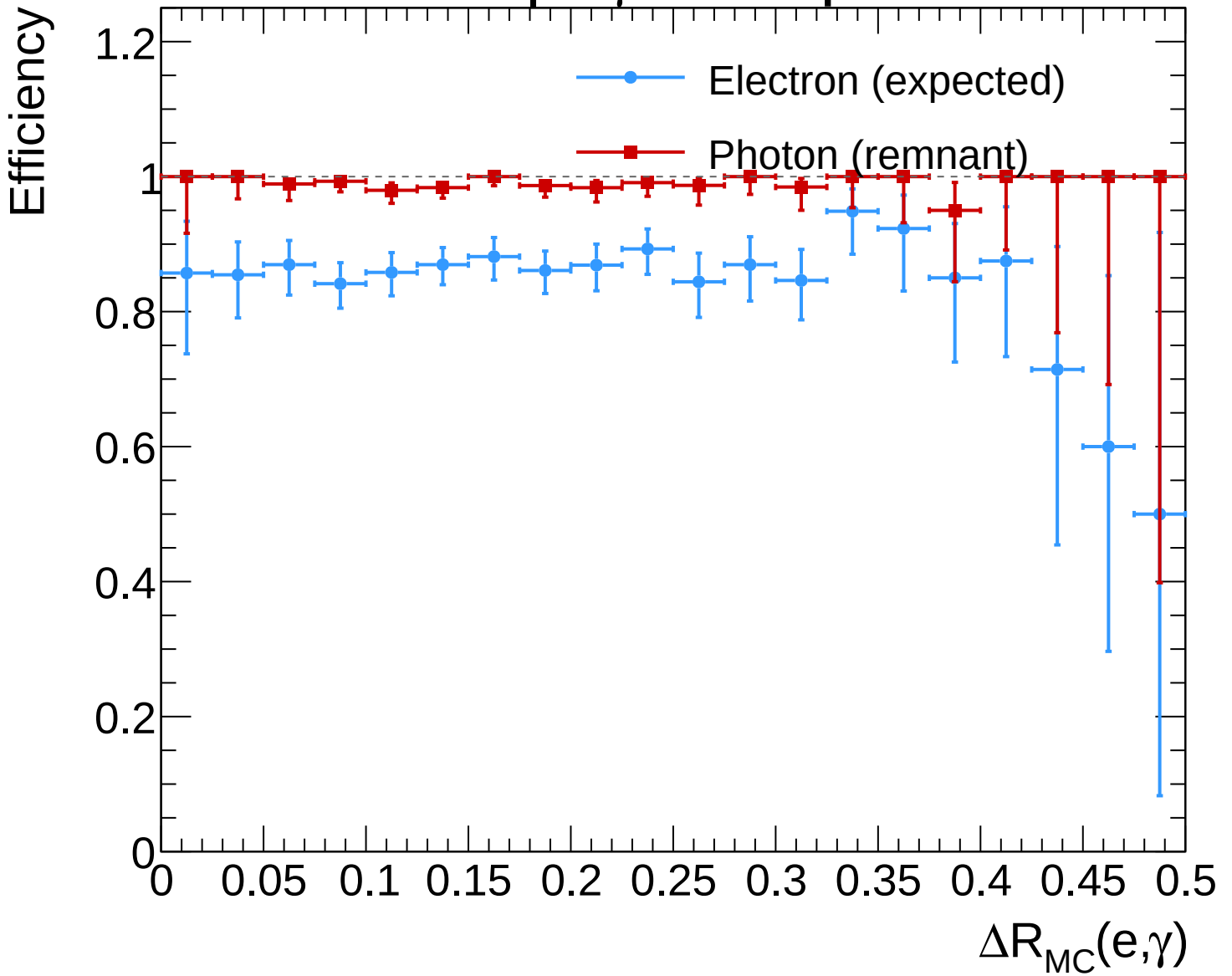

ePIC Benchmark | e⁻+γ N Endcap ECal
Black: mean profile
E_exp / E_MC,e
ΔR_MC(e,γ)
0
0.2
0.4
0.6
0.8
1
1.2
1.4
1.6
1.8
2
0.05
0.1
0.15
0.2
0.25
0.3
0.35
0.4
0.45
0.5
5
10
15
20
25
30

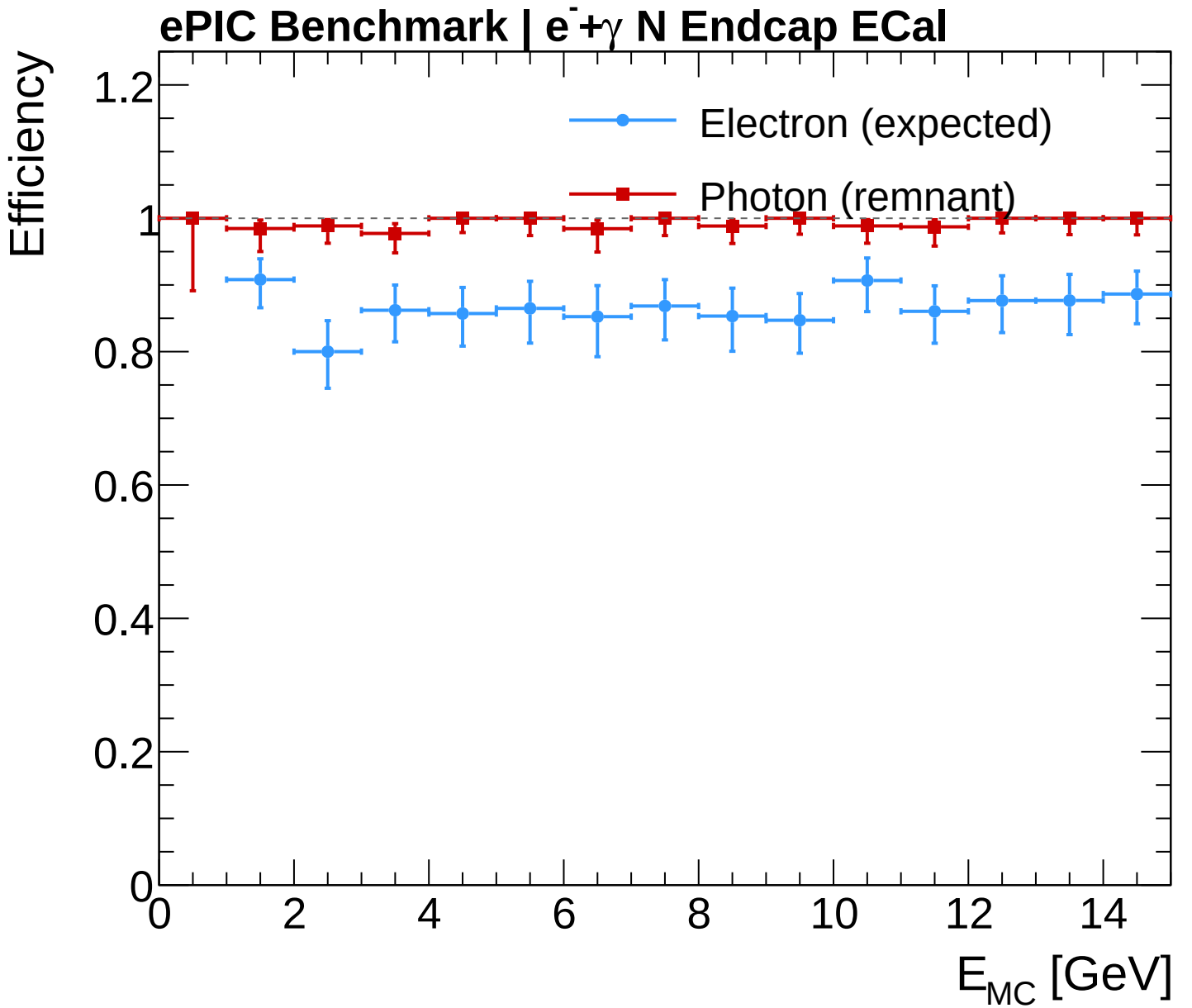

ePIC Benchmark | e⁻+γ N Endcap ECal
Black: mean profile
ΣE_rem / E_MC,γ
ΔR_MC(e,γ)
0
0.2
0.4
0.6
0.8
1
1.2
1.4
1.6
1.8
2
0.05
0.1
0.15
0.2
0.25
0.3
0.35
0.4
0.45
0.5
5
10
15
20
25

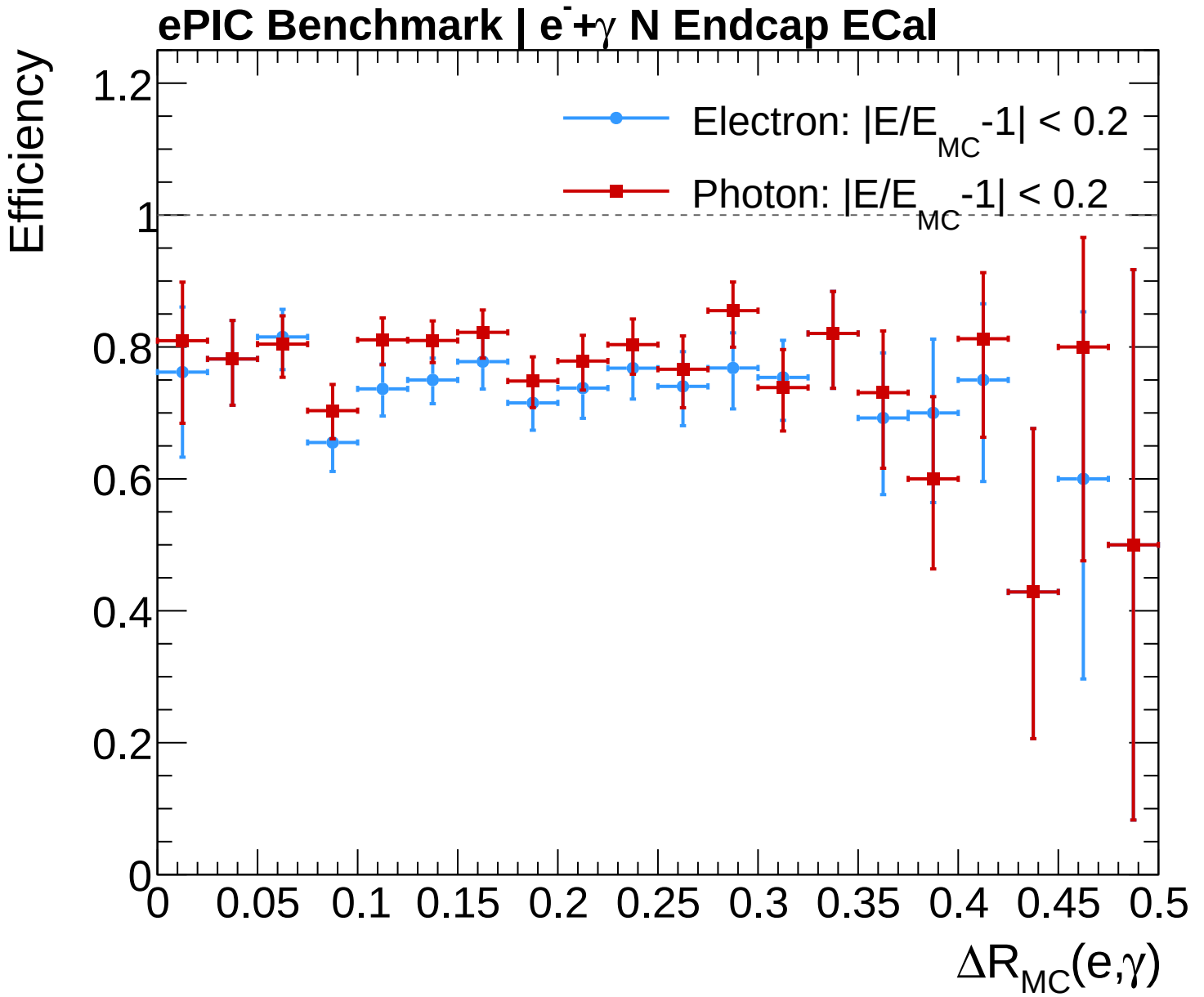

ePIC Benchmark | e⁻+γ N Endcap ECal
< E_reco / E_MC >
Electron: E_exp/E_MC,e
Photon: ΣE_rem/E_MC,γ
ΔR_MC(e,γ)
0
0.2
0.4
0.6
0.8
1
1.2
1.4
1.6
0.05
0.1
0.15
0.2
0.25
0.3
0.35
0.4
0.45
0.5


ePIC Benchmark | e⁻+γ N Endcap ECal
< E_reco,tot / E_MC,tot >
(E_exp+ΣE_rem) / (E_MC,e+E_MC,γ)
ΔR_MC(e,γ)
0
0.2
0.4
0.6
0.8
1
1.2
1.4
1.6
0.05
0.1
0.15
0.2
0.25
0.3
0.35
0.4
0.45
0.5


ePIC Benchmark | e⁻+γ N Endcap ECal
< E_exp / p_Tr >
E_exp / p_Tr
1.6
1.4
1.2
1
0.8
0.6
0.4
0.2
0
0
0.05
0.1
0.15
0.2
0.25
0.3
0.35
0.4
0.45
0.5
ΔR_MC(e,γ)


ePIC Benchmark | e⁻+γ N Endcap ECal
Efficiency
Electron (expected)
Photon (remnant)
1.2
1
0.8
0.6
0.4
0.2
0
0
0.05
0.1
0.15
0.2
0.25
0.3
0.35
0.4
0.45
0.5
ΔR_MC(e,γ)


ePIC Benchmark | e⁻+γ N Endcap ECal
Efficiency
1.2
1
0.8
0.6
0.4
0.2
0
0
2
4
6
8
10
12
14
E_MC [GeV]
Electron (expected)
Photon (remnant)


ePIC Benchmark | e⁻+γ N Endcap ECal
Efficiency
Electron: |E/E_MC -1| < 0.2
Photon: |E/E_MC -1| < 0.2
ΔR_MC(e,γ)
0
0.2
0.4
0.6
0.8
1
1.2
0.05
0.1
0.15
0.2
0.25
0.3
0.35
0.4
0.45
0.5